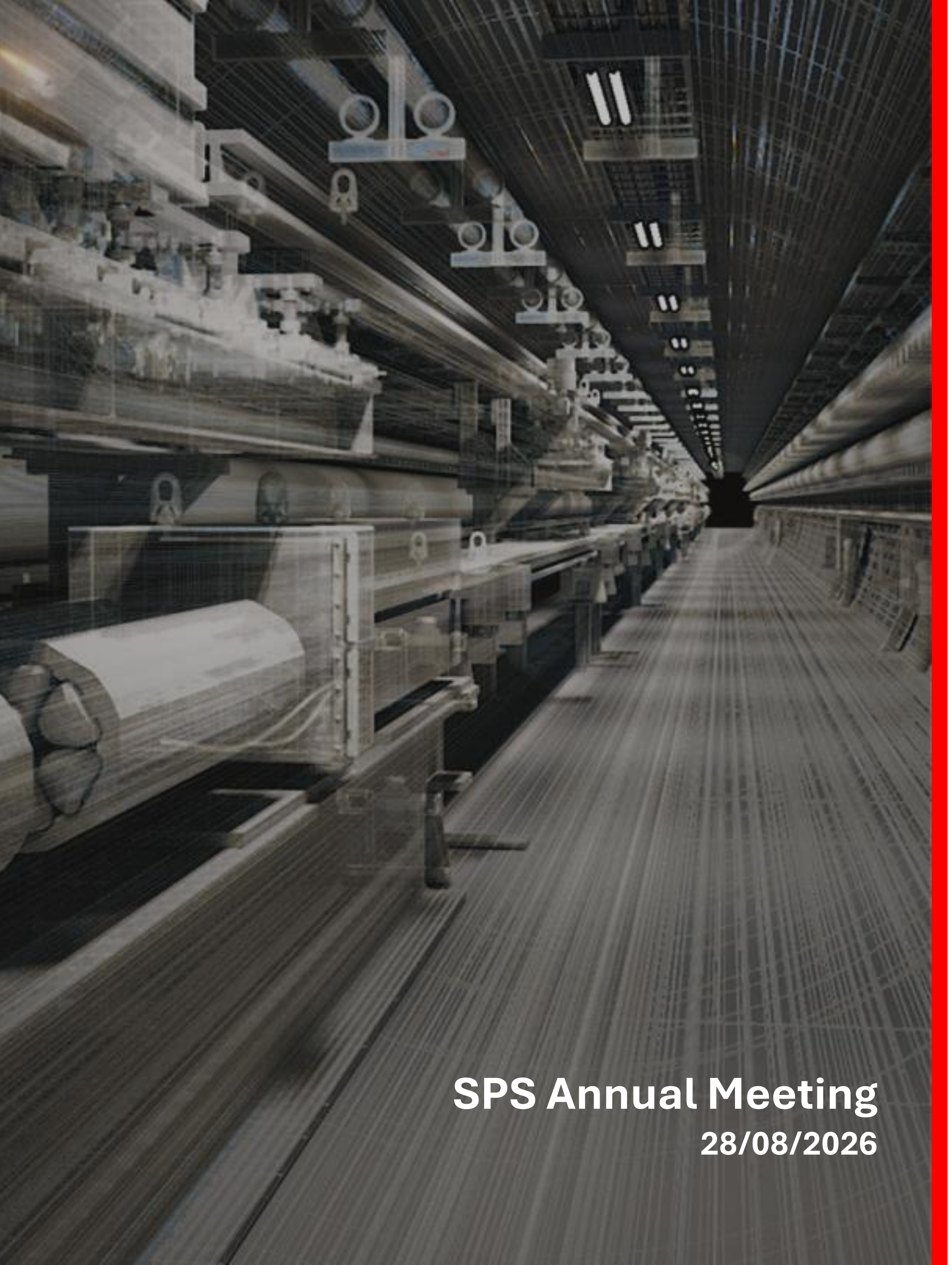

SPS Annual Meeting
28/08/2026

FUTURE CIRCULAR COLLIDER

CHART Swiss Accelerator Research and Technology

EPFL

Part I: FCC-ee Studies with Imperfections and Beam-Beam Effects

Lara Sievert

Under the supervision of T. Prebibaj, T. Pieloni, M. Seidel

With the help and support of:
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K. Skoufaris, P. Raimondi, L. Van Riesen-Haupt, Y. Wu

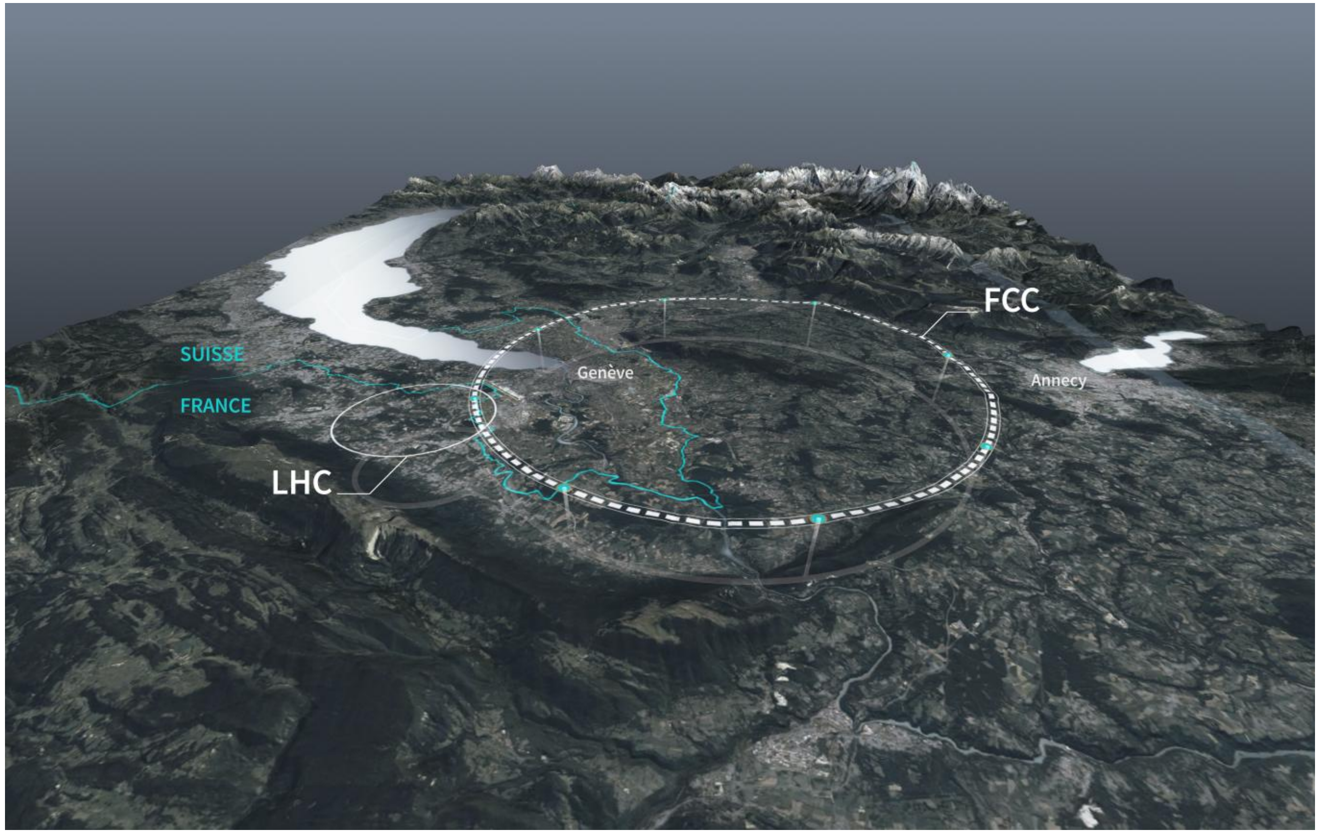
Work supported by the Swiss Accelerator Research and Technology (CHART)

École polytechnique fédérale de Lausanne

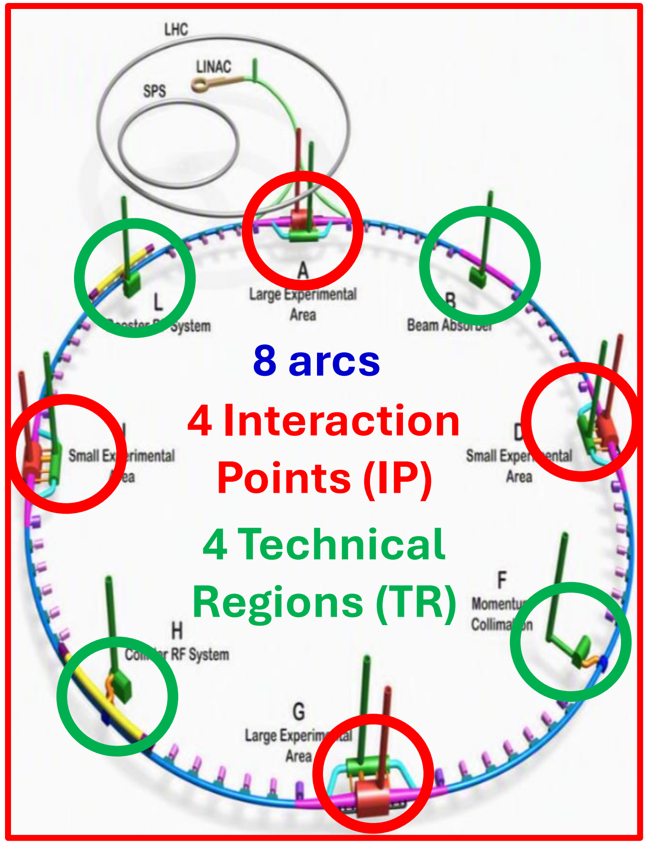
Outline

I. Introduction

Overview of FCC-ee



Source: CERN Future Circular Collider



Adapted from FCC Feasibility Study Report.

The Future Circular electron-positron Collider (FCC-ee):

- Proposed next-generation particle collider of 90.7 km length
- Designed to operate at four energies: 45.6 GeV (Z), 80 GeV (WW), 120 GeV (ZH), and 182.5 GeV ($t\bar{t}$)

FUTURE CIRCULAR COLLIDER

Overview of Important Definitions

In a simplified fashion...

The **accelerator lattice** consists of different magnets:
dipoles for beam bending, quadrupoles for beam focusing, and higher-order magnets (e.g. sextupoles)

ORBIT:

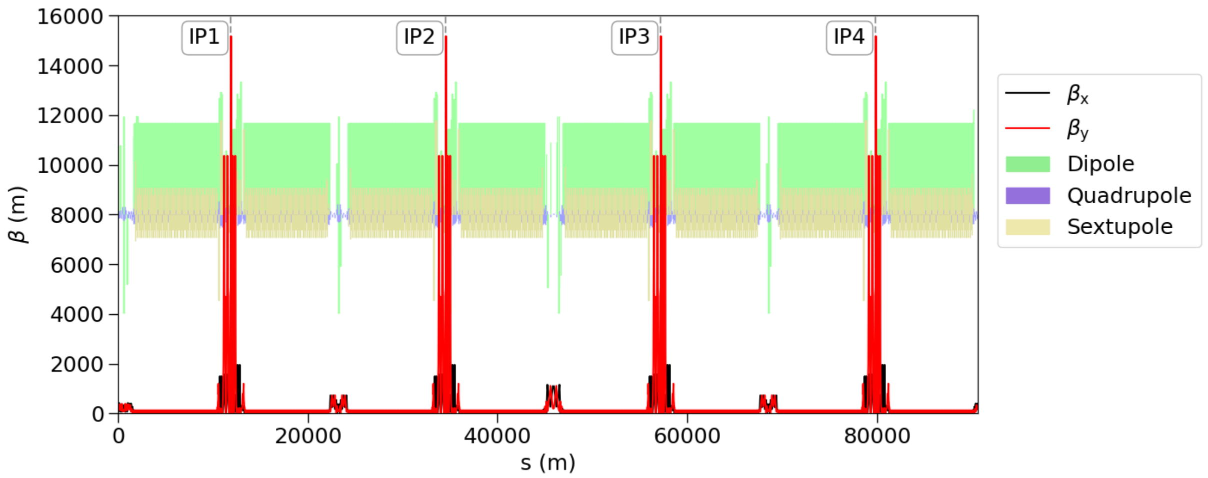
average beam trajectory around the ring, affected mainly by dipoles

OPTICS:

set of parameters characterising beam size & motion (betatron oscillations), affected mainly by quadrupoles

- **Beta-function β:** linked to the beam size
- **Dispersion D:** how energy spread affects the beam motion
- **Coupling:** how motion in one transverse plane is affected by the other plane
- **Tune:** number of betatron oscillations per revolution

In an ideal case, the magnets are all in the right position
→ see Twiss in the ideal lattice scenario



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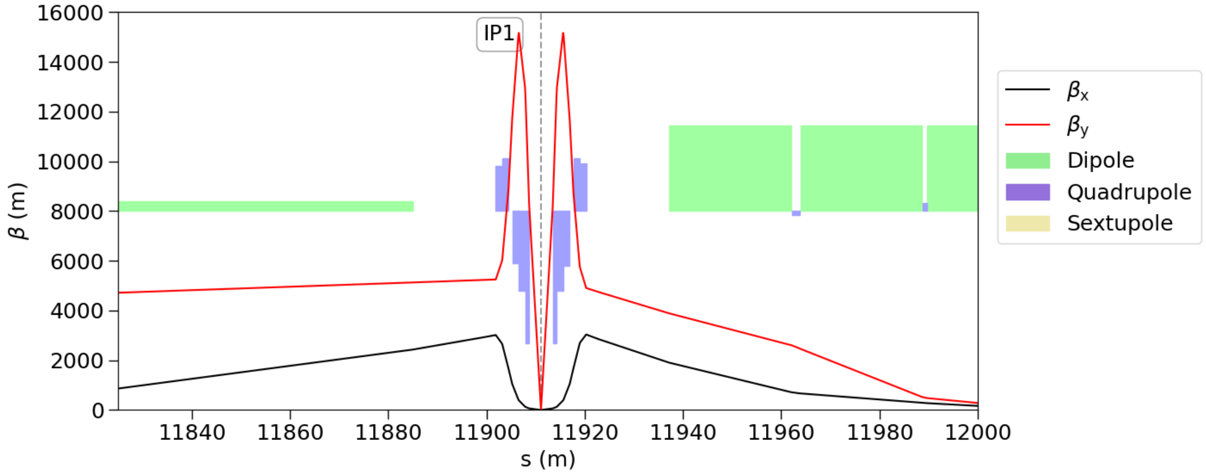
But in the real world, we need to consider **imperfections**...



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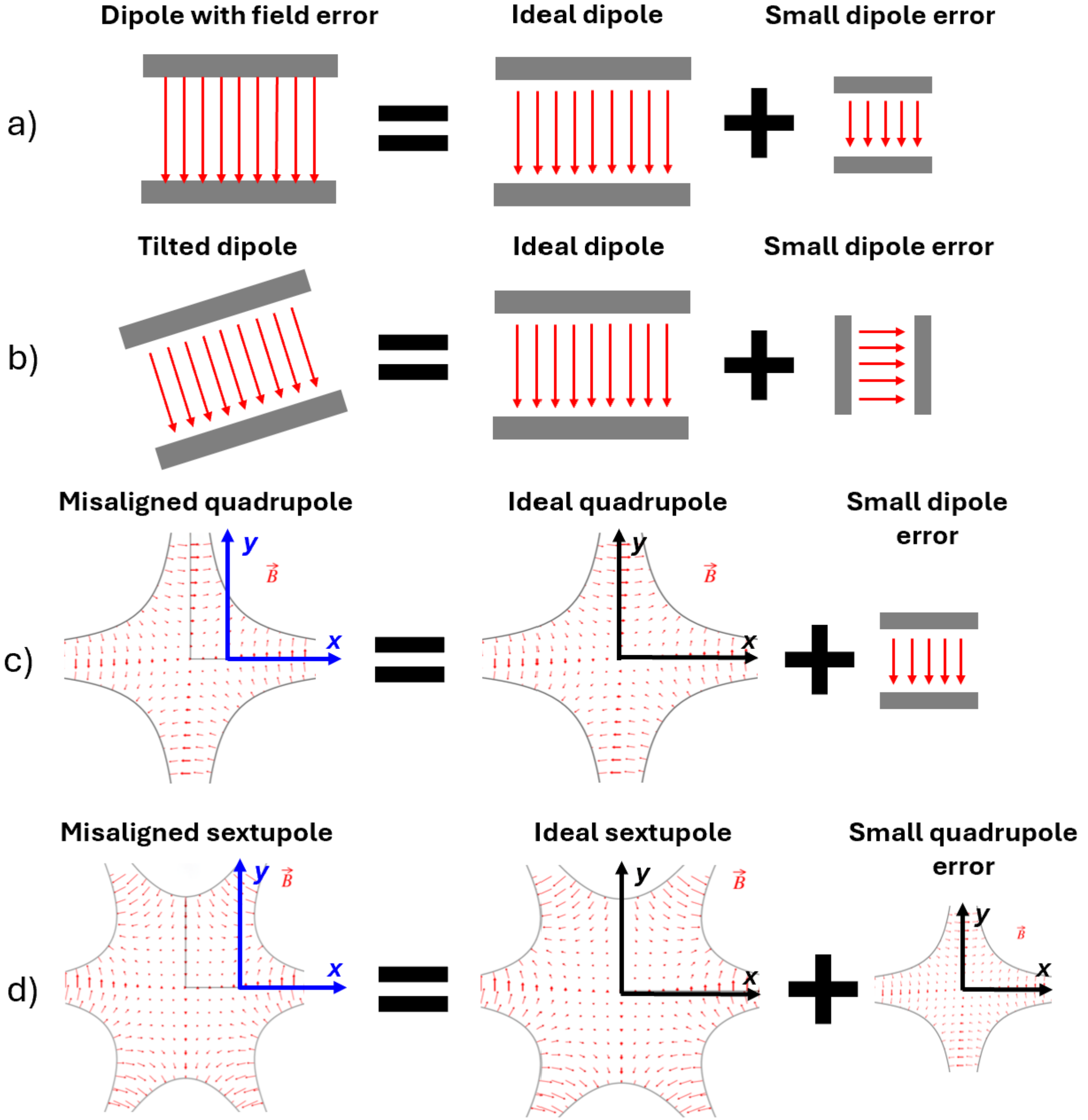
Overview of Imperfections

	Field type	Imperfection	Error type	Impact
a)	Dipole	Field error	Dipole	Orbit, trajectory, energy
b)	Dipole	Roll	Dipole	Orbit, trajectory
	Quadrupole	Field error	Quadrupole	Tune, optics
c)	Quadrupole	Offset	Dipole	Orbit, trajectory
	Quadrupole	Roll	Skew quadrupole	Coupling
	Sextupole	Field error	Sextupole	Chromaticity
d)	Sextupole	Horizontal offset	Quadrupole	Tune, optics
	Sextupole	Vertical offset	Skew quadrupole	Coupling

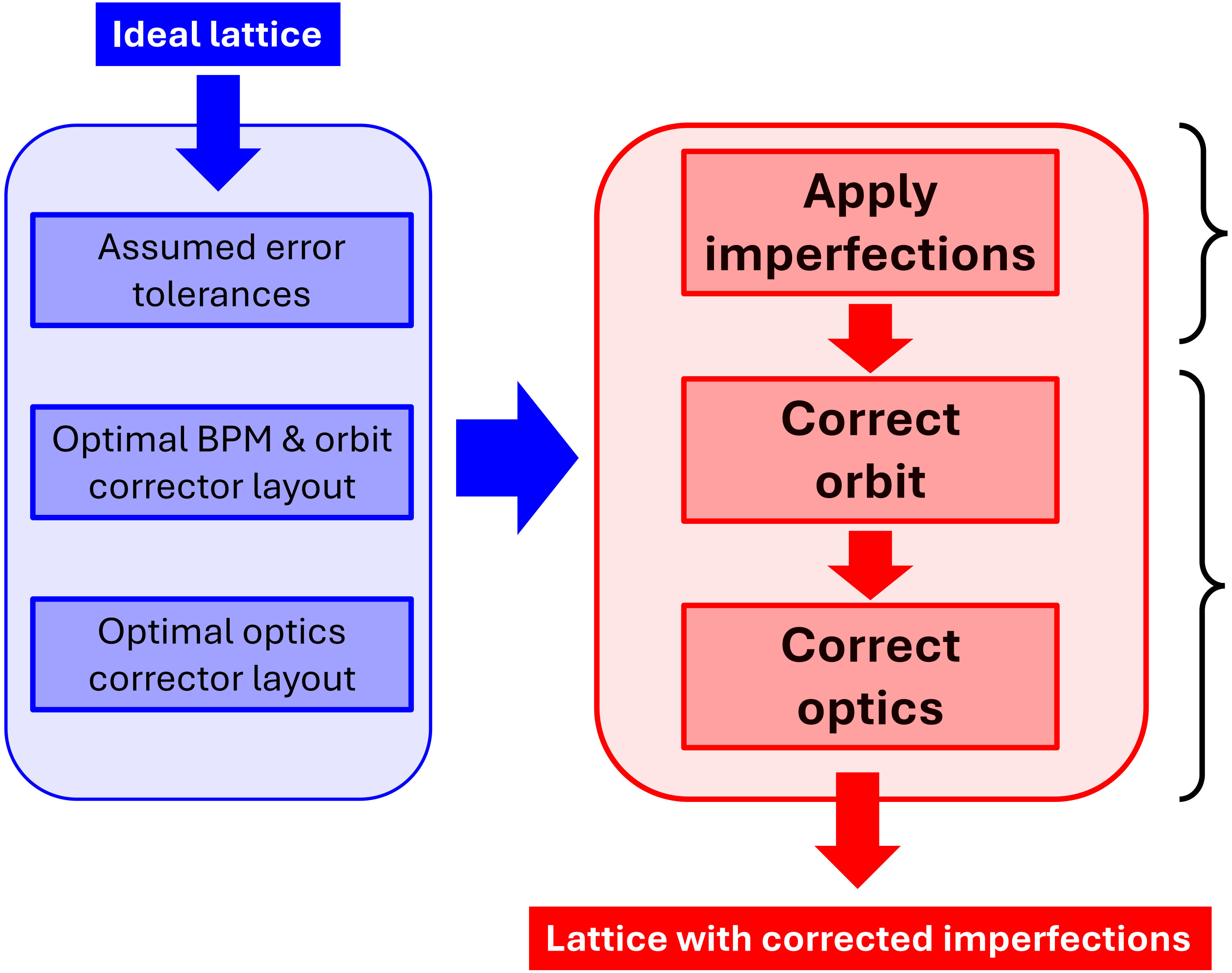
J. Wenninger, "Linear Imperfections", CAS 2020.



Adapted from J. Wenninger.

II. Corrected Lattices

From the ideal lattice to a more realistic lattice



- Install misalignments, rotations, field errors, …
- Based on the assumed tolerances for elements in the arcs, IRs, TRs

- Use the response matrix approach
- To recover a closed orbit & correct for orbit excursions
- To correct for distortions in the β-function, dispersion, coupling, etc.

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Orbit Correction Basics

Similar approach for **optics correction**!

Aim: Compensating for the unintended distortions caused by lattice imperfections by using dedicated corrector magnets

→ **Response matrix approach**

$$\Delta u_i = R_{ij}\,\Delta\theta_j$$

Orbit variation measured at BPM i

Response matrix (RM)

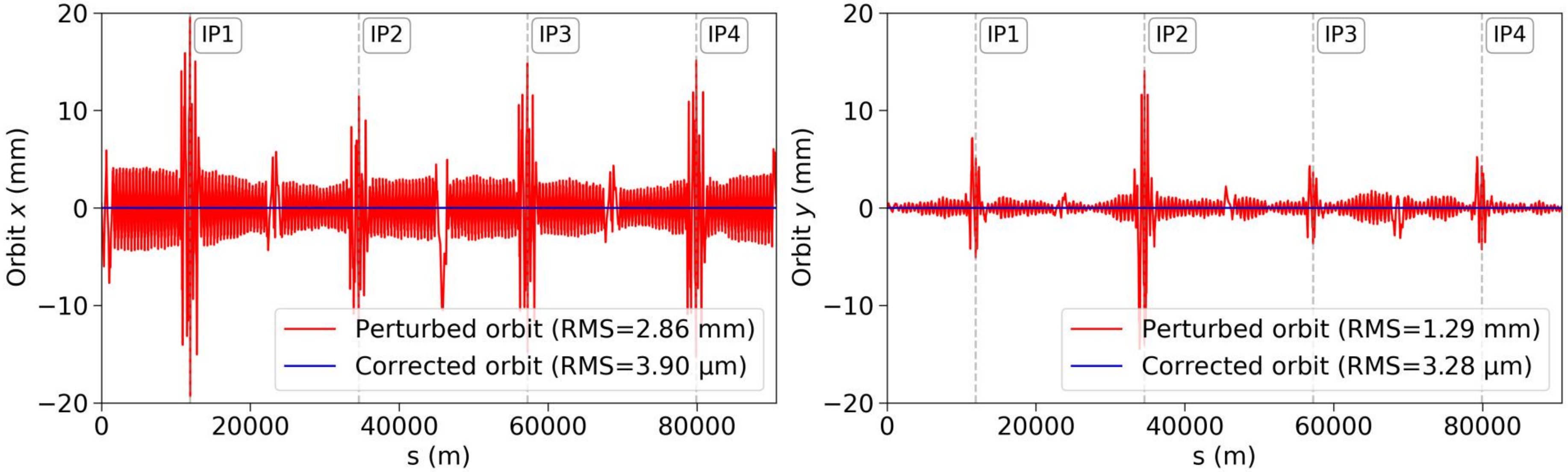
Kick applied by corrector j

Pseudo-inversion of RM using SVD: $\Delta\vec{\theta} = R^{\dagger}\,\Delta\vec{u}$

Orbit correction steps

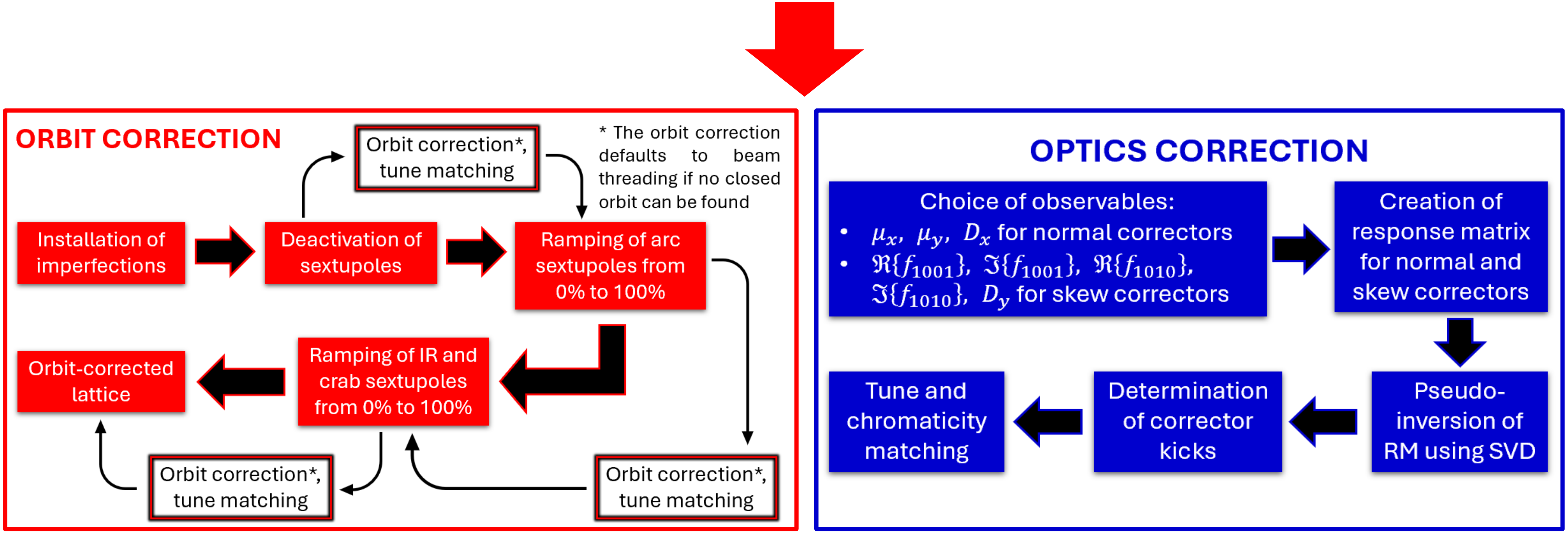
1. Construction of orbit RM
2. Pseudo-inversion via singular value decomposition (SVD)
3. Application of the required corrector strengths to modify the beam trajectory

In practice: Correcting the orbit in a lattice with arc quadrupole misalignments



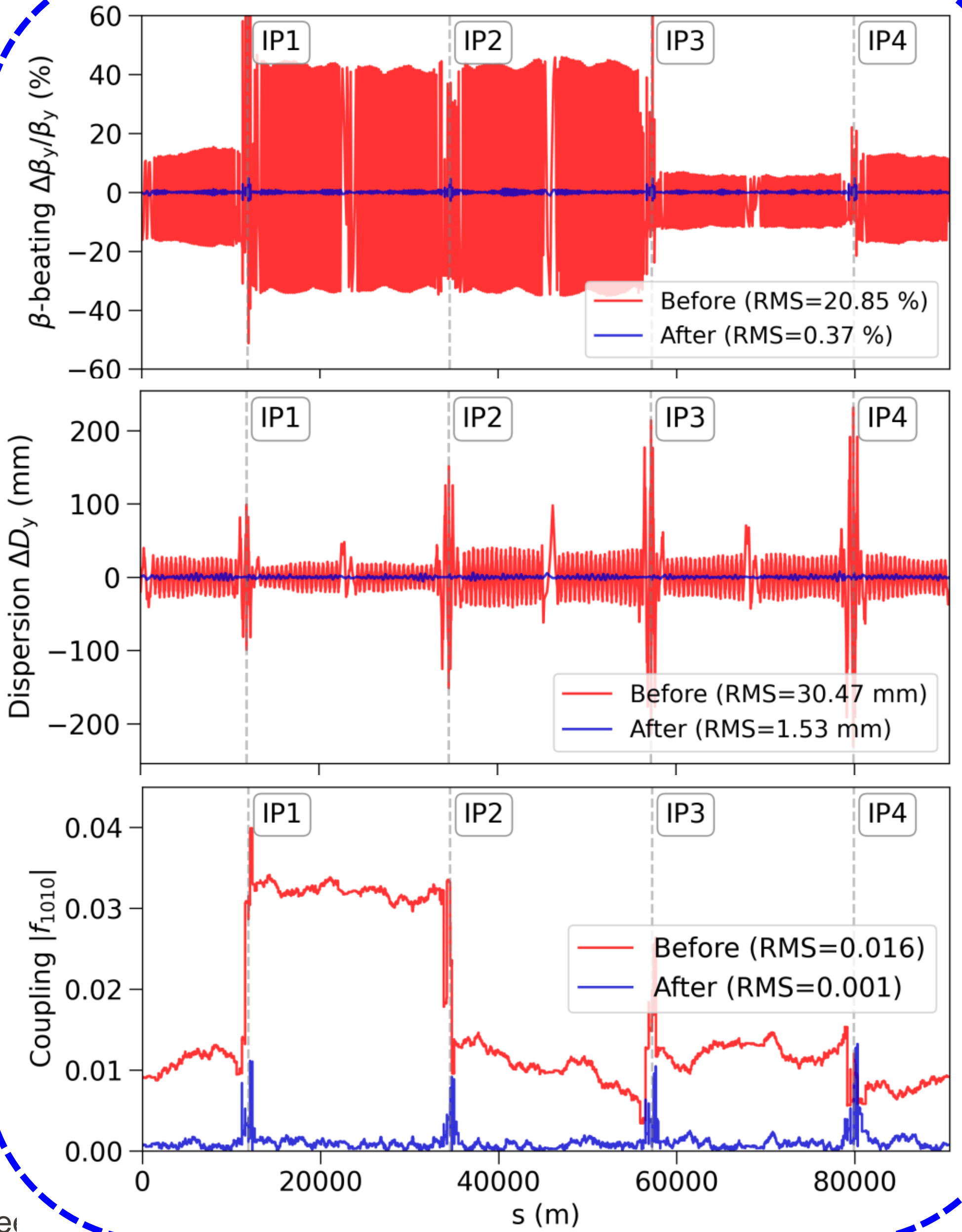
Orbit and Optics Correction Routine

Applying imperfections (misalignments, rotations, field errors) to all dipoles, quadrupoles, sextupoles in the lattice



Correction Levels

Repeated over many “seeds”

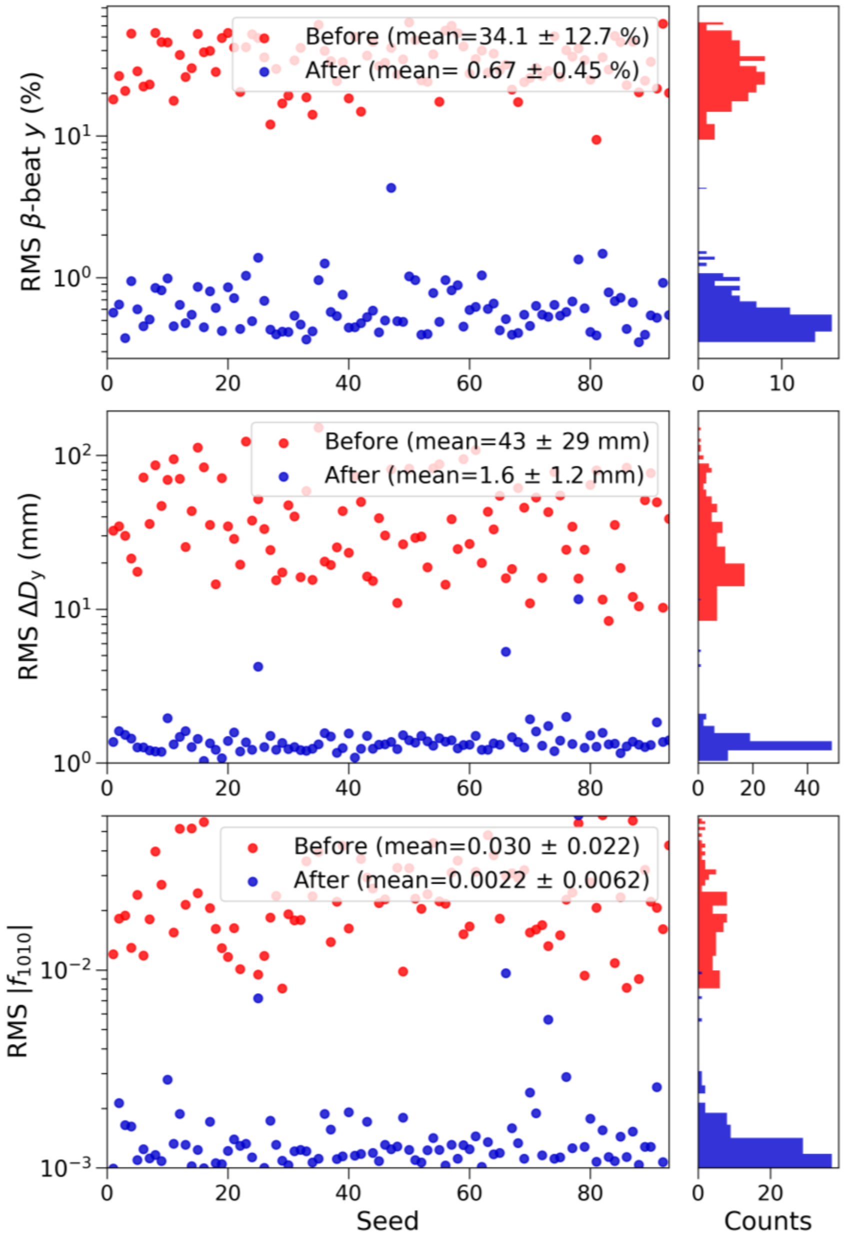


β-beating

Dispersion

Coupling

Statistics results for 100 seeds



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Correction Levels

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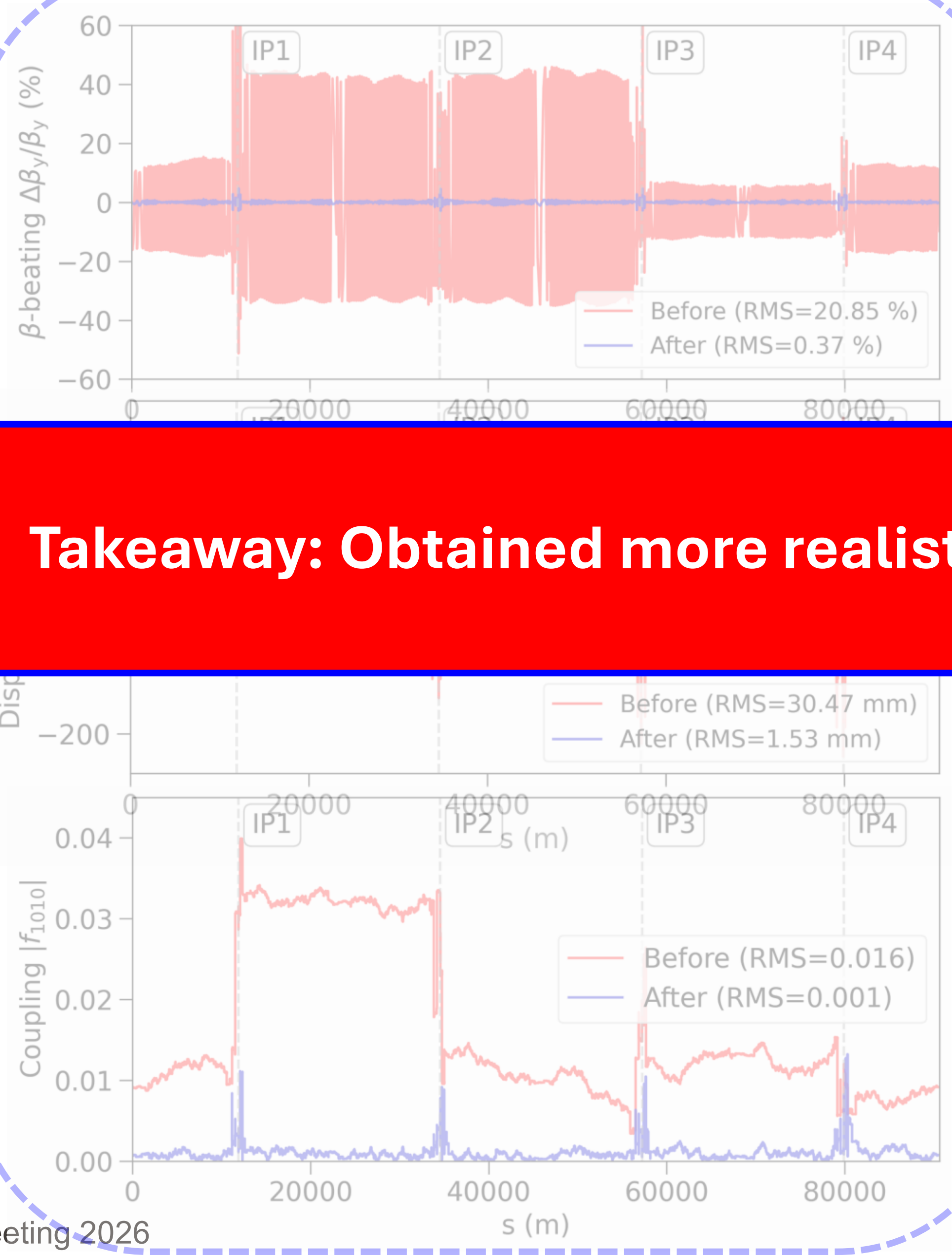
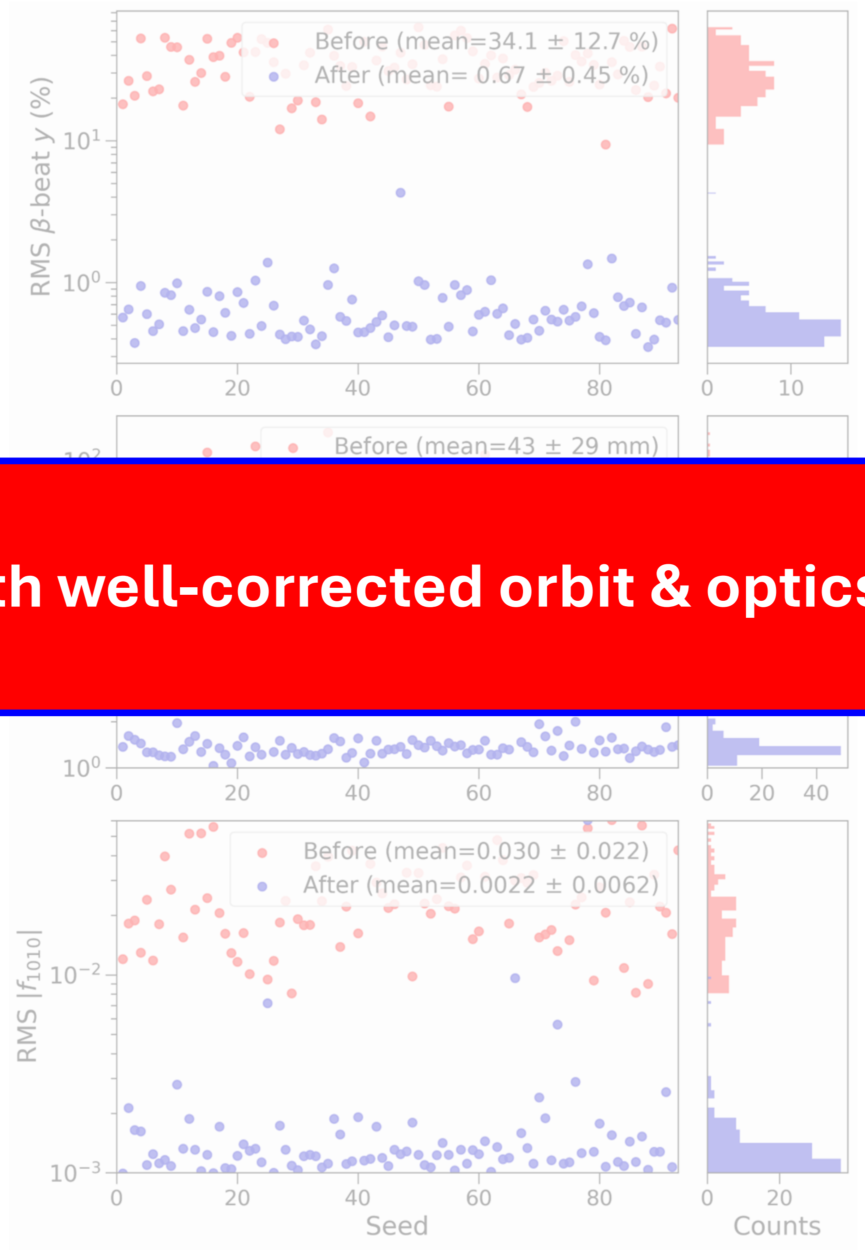


β-beating

Coupling

Statistics results for 100 seeds



Takeaway: Obtained more realistic lattices with well-corrected orbit & optics.

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Using the corrected lattices for beam-beam studies

Ideal lattice

Assumed error tolerances

Optimal BPM & orbit corrector layout

Optimal optics corrector layout

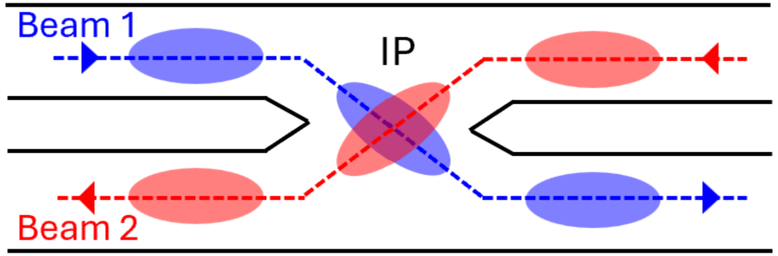
Apply imperfections

Correct orbit

Correct optics

Lattice with corrected imperfections

Install beam-beam effects



To study the interplay with the residual orbit and optics distortions after correction

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III. Beam-Beam Studies

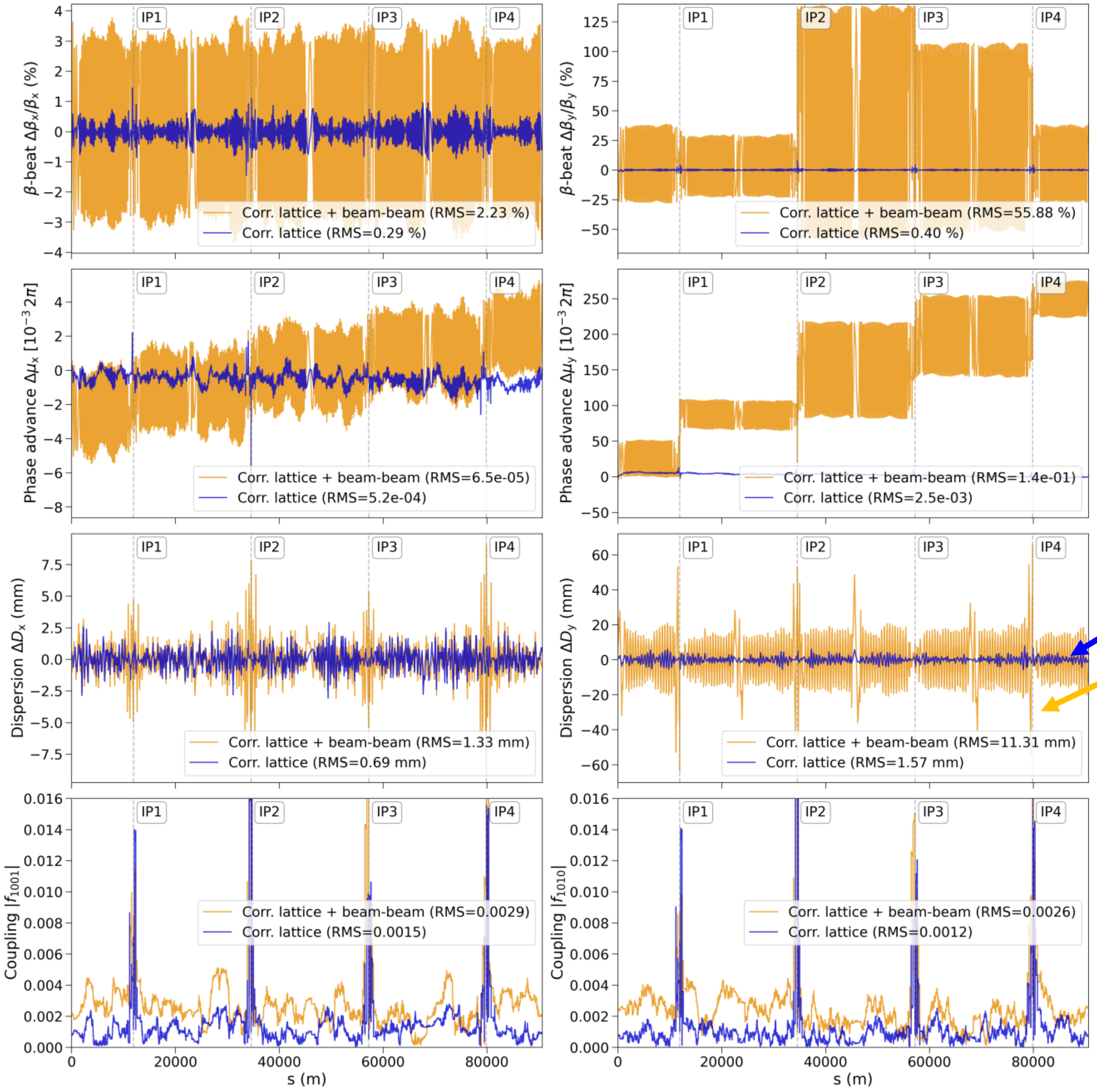
Beam-Beam and Impact on Optics

β-beating

Phase adv.

Dispersion

Coupling



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
  - → e.g. see IPAC paper[1]

Corrected lattice

Corrected lattice with beam-beam effects

Beam-beam parameter

$\xi_x = 0.0014$ and $\xi_y = 0.09$

[1] T. Prebibaj, L. Sievert, L. van Riesen-Haupt, M. Seidel, P. Raimondi, T. Pieloni, and Y. Wu, "Simulations of Phase-Advance Correction for FCC-ee and Impact on its Perfor mance." presented at IPAC'26, Deauville, France, paper MOP1077, to be published, 05 2026.

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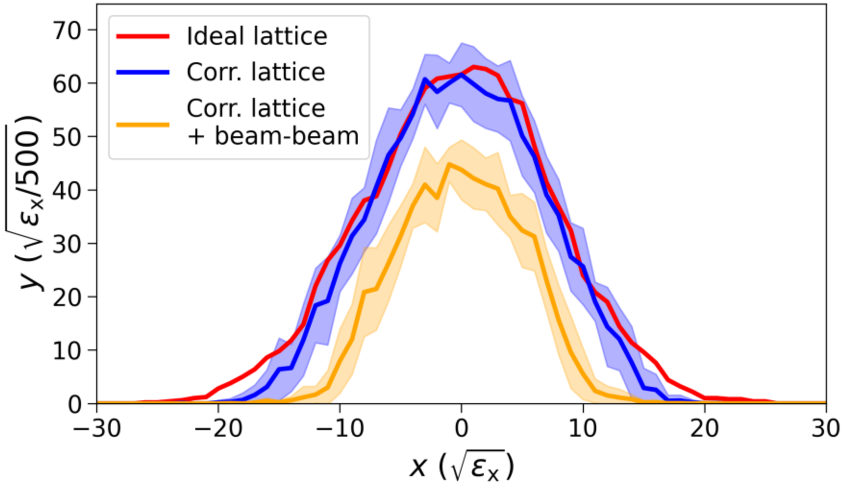
Beam-Beam and Impact on Performance

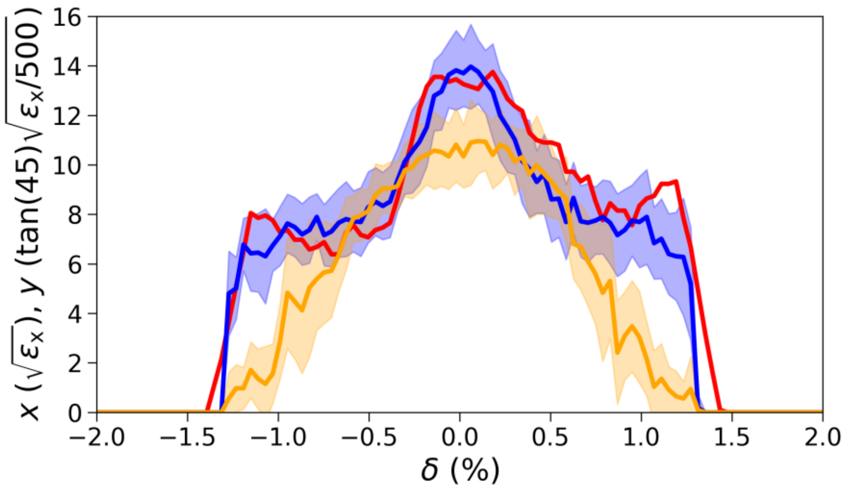
Dynamic Aperture / Momentum Acceptance

"Region in which particles can stably circulate"

DA



MA



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
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- Quantifying beam-beam induced **performance degradation** in lattices with imperfections
  → e.g. see L. Sievert's Master's thesis defence[2]

[2] L. Sievert, "Master's Thesis Defence: Imperfections Model of the FCC-ee with Corrections and Beam-Beam Effects." EPFL, 07 2026.

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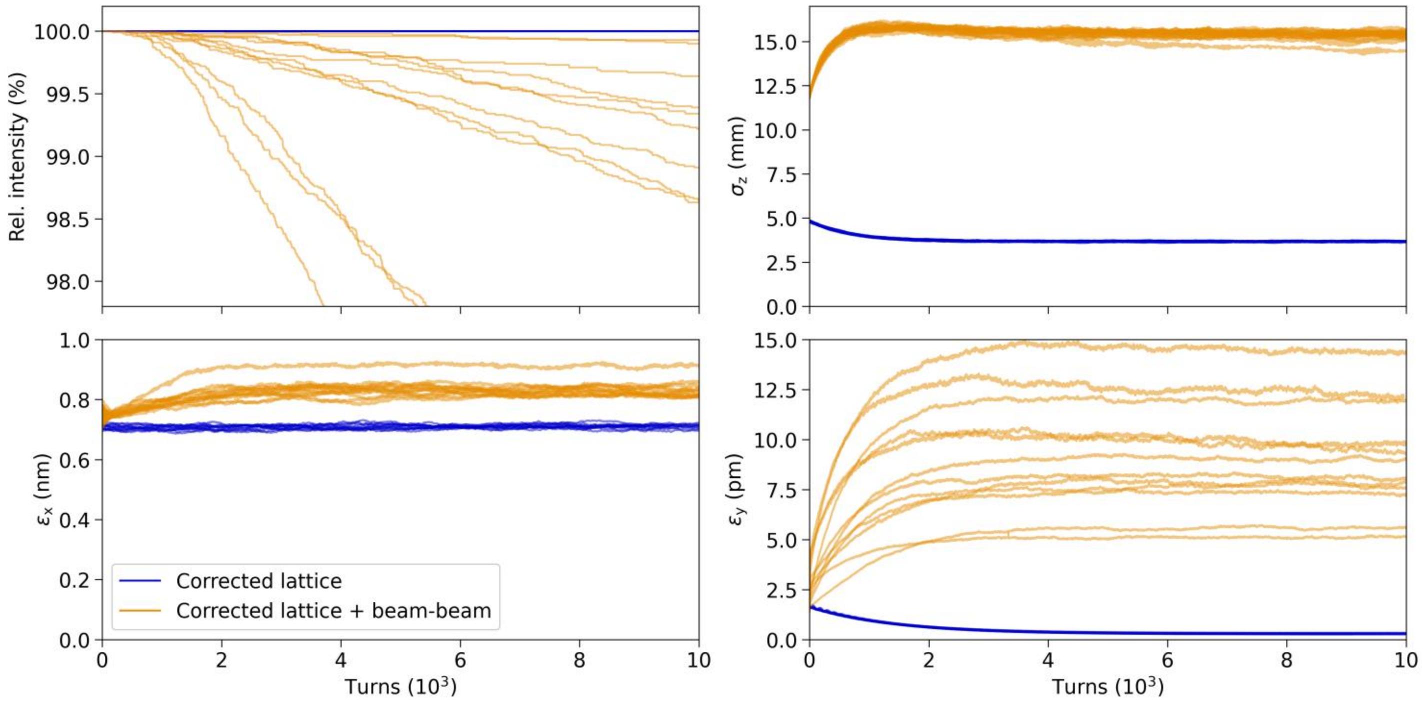
Beam-Beam and Impact on Performance

Tracking results



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
  → e.g. see IPAC paper
- Quantifying beam-beam induced **performance degradation** in lattices with imperfections
  → e.g. see L. Sievert's Master's thesis defence[2]

Details on simulation setup

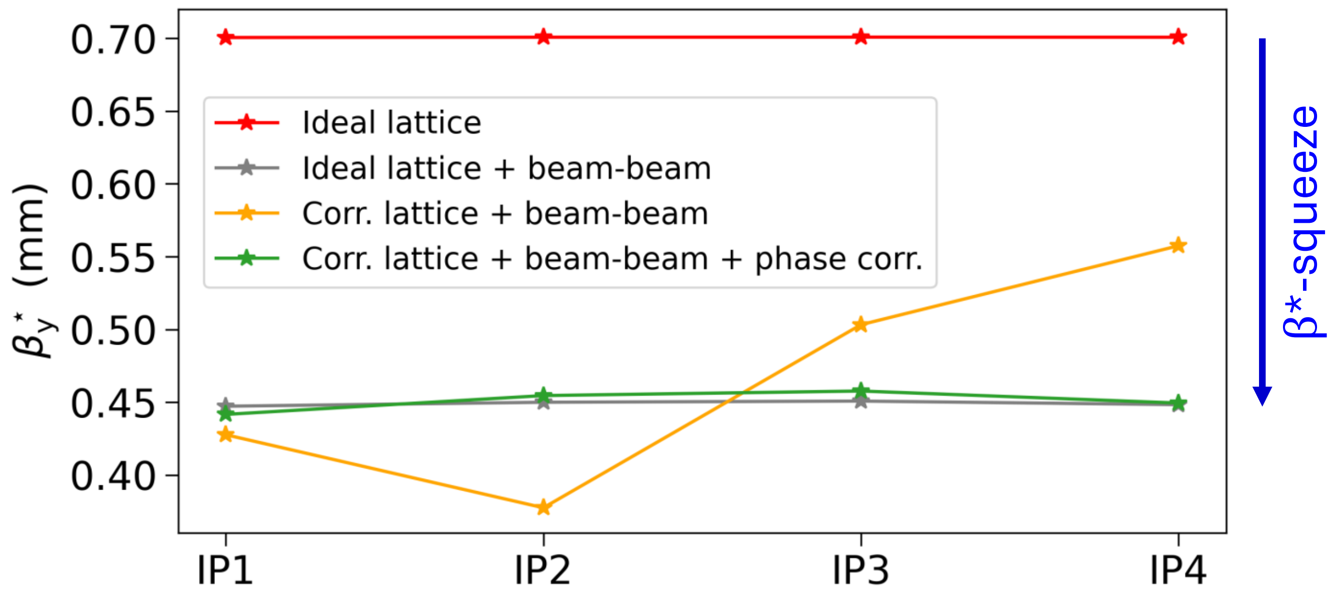
- Tracking 10^4 particles for 10^4 turns in the corrected lattices
- Beam-beam interactions are modelled using the "quasi-strong-strong" scheme
- Synchrotron radiation with quantum excitations, Beamstrahlung, and tapering are included

[2] L. Sievert, "Master's Thesis Defence: Imperfections Model of the FCC-ee with Corrections and Beam-Beam Effects." EPFL, 07 2026.

Mitigating beam-beam induced optics degradations

- On top of the global optics correction, local corrections are needed at the IPs (**IP tuning**) to target the beam-beam induced distortions
  → e.g. **phase correction** to match the inter-IP phase advance back to the nominal value in corrected lattices with beam-beam
- Impact on the optics of IP tuning in corrected lattices with beam-beam to be evaluated

Example: **Restoring β^* symmetry between IPs**



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
  → e.g. see IPAC paper
- Quantifying beam-beam induced **performance degradation** in lattices with imperfections
  → e.g. see L. Sievert's Master's thesis defence
- Developing techniques for recovering optics and performance with beam-beam
  → e.g. see T. Prebibaj's FCC week presentation[3] & upcoming talk "IP Tuning with Beam-Beam"

[3] T. Prebibaj, "FCC week presentation: Beam-beam interplay with optics and tuning." presented at FCC week '26, Helsinki, Finland, 06 2026.

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Conclusions

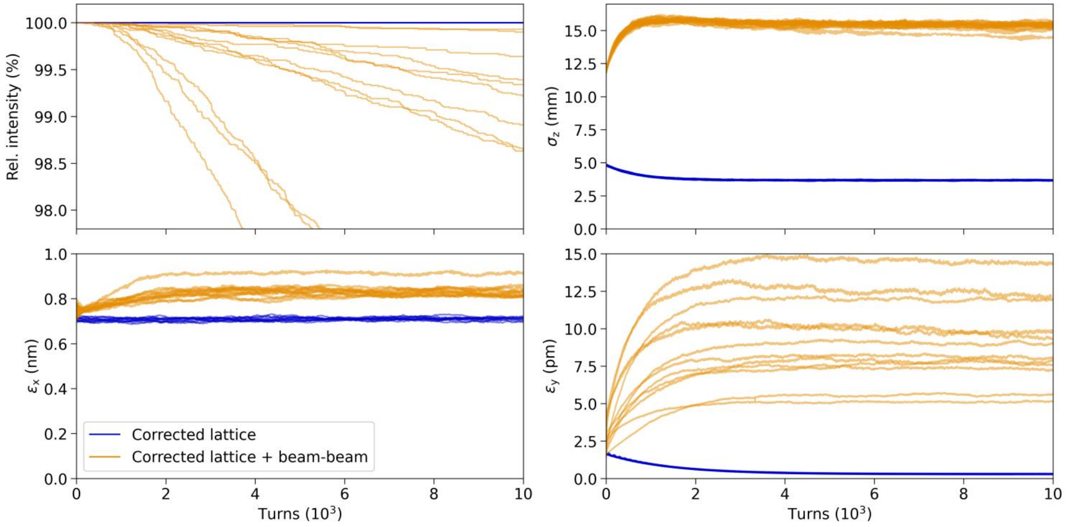
Takeaways

- **Developed model for lattice imperfections with orbit and optics corrections to study beam-beam in more realistic lattices**

  → Developed in Xsuite for FCC-ee at Z, but generic implementation of model allows for it to be extended to other machines
  → Corrected lattices all available on Github: ***https://github.com/Sievert-L/FCCee_Project_2026***

- **Observed significant beam-beam induced degradations in corrected lattices**

  → Quantified impact on optics and on collider performance
  → Mitigation methods needed to restore performance: IP tuning (see next talk by Tirsi Prebibaj)



Thank you for your attention!

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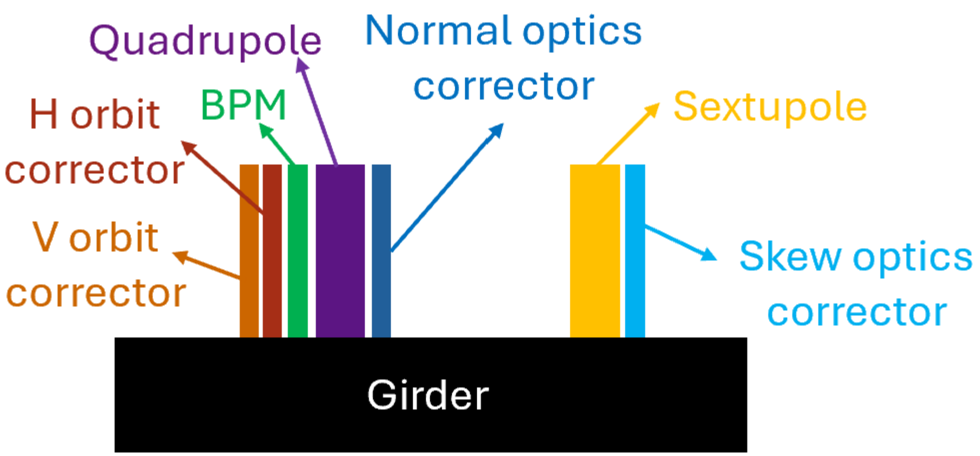
Supporting Material

Details on lattice layout

Element type	# Elements
Dipole	2432
Quadrupole	2679
Sextupole	1952
BPM	2639
H / V orbit corrector	2679
Normal optics corrector	2663
Skew optics corrector	1944



Schematic of an arc section (not to scale)

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Assumed error tolerances

Section	Element	Misalignment			Rotation	Field error (rel.)
		σ_x (μm)	σ_y (μm)	σ_s (μm)	σ_θ (μrad)	
ARC	Dipole	1000	1000	500	1000	k_0: 1×10^{-3}
	Quadrupole	50	50	100	50	k_1: 2×10^{-4}
	Sextupole	50	50	100	50	k_2: 2×10^{-4}
	Girder	150	150	500	150	–
STRAIGHT SECTION	IR Dipole	1000	1000	100	1000	k_0: 1×10^{-3}
	TR Dipole	1000	1000	500	1000	k_0: 1×10^{-3}
	FD Quadrupole	30	30	100	30	k_1: 5×10^{-5}
	FF Quadrupole (excl. FD)	30	30	100	30	k_1: 1×10^{-4}
	TR Quadrupole	100	100	100	100	k_1: 2×10^{-4}
	FF Sextupole (incl. CS)	30	30	100	30	k_2: 1×10^{-4}
–	BPM	10	10	–	100	–

IR = Interaction Region

TR = Technical Region

FD = Final Doublet

FF = Final Focussing

CS = Crab Sextupole

- Imperfections include:
  - **Misalignments** in transverse and longitudinal planes (x, y, s)
  - **Rotations** around the longitudinal axis (s-rotation)
  - **Field errors**
  - **BPM displacement errors**
- Imperfections are applied randomly based on a Gaussian distribution truncated at 2.5σ → different "seeds"
- BPMs inherit imperfections from quadrupoles, which themselves may inherit from girder
- Not yet included: Higher-order errors, systematic errors, measurement errors

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Tuning results

Observable	LCC Z-pole (V106.2.3)
failed seeds	7 / 100
rms orbit H / V (μm)	137 / 138
rms H / V orbit corrector strength in ARC (μrad)	11.55 / 11.63
rms H / V orbit corrector strength in IR (μrad)	2.23 / 2.11
rms H / V orbit corrector strength in TR (μrad)	4.66 / 4.67
rms normal / skew optics corrector strength (mT)	0.77 / 0.50
rms $\Delta D_{x/y}$ (mm)	0.66 / 1.55
rms $\Delta\mu_{x/y}$ $[10^{-4}\ 2\pi]$	13.2 / 27.3
rms $\Delta\beta_{x/y}/\beta_{x/y}$ (%)	0.55 / 0.67
rms $\lvert f_{1001}\rvert/\lvert f_{1010}\rvert$ $[10^{-4}]$	26.4 / 21.8
median ε_x (nm) / ε_y (pm) [from Twiss method]	0.71 / 0.028
transverse tunes Q_x/Q_y	194.16 / 170.20
rms $\Delta D^{(2)}_{x/y}$ (mm)	363 / 635

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